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RESEARCH ARTICLE

DermaTransNet: Where Transformer Attention Meets U-Net for Skin Image Segmentation

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ABSTRACT Skin, the largest organ of the human body, serves as a protective barrier against the external environment, safeguarding the internal organs. It is composed of layers, namely the epidermis, dermis, and hypodermis. The analysis and detection of diseases heavily rely on identifying the penetration of biomarkers within each layer, thus highlighting the significance of accurate segmentation of each layer in disease diagnosis. In skin disease diagnosis, haematoxylin and eosin-stained whole slide images are the most commonly employed imaging modality. To enhance the efficacy of analysis and assist pathologists, computer-aided diagnostic systems have been proposed to automate the process of disease detection and classification. A crucial initial step in skin disease diagnosis is the accurate segmentation of skin layers, as it facilitates the identification of regions of interest for subsequent processing. This study presents a novel attention-based encoder-decoder architecture comprising a multi-axis encoder with an attention-mixing decoder to precisely segment the skin layers (Epidermis, Dermis, Hypodermis, Keratin) from stained whole slide image samples. The proposed Transformer-based encoder block has helped the model preserve global and local features, which are then passed to the decoder block through an attention-based gated skip connection. The attention mixing decoder module has combined multi-head self-attention, spatial attention, and squeeze excitation modules to enhance the spatial information gain. Additionally, the encoder block aids in extracting high and low-level features, which are further weighted before passing to the decoder block via a gated attention mechanism, thus helping the model acquire the relevant details from all spatial dimensions. The proposed architecture has been evaluated on a publicly available dataset against various evaluation metrics i.e., F1-Score, Accuracy, Precision, Recall, and Hausdorff Distance. The model yielded the highest accuracy of 95.5% when compared with state-of-the-art segmentation models. Additionally, compared with the baseline framework, the model has resulted in an increased overall accuracy of 14.20%.

INDEX TERMS Histopathology, whole slide image, semantic segmentation, transformer, skin layers, histopathology, computer-aided diagnostics, bright field microscopy, UNet architecture, attention.

I. INTRODUCTION

The human integumentary system [1] is primarily composed of skin, which provides a protective layer between the external environment and the human body. Moreover, it is the largest organ, weighing one-seventh of the total body weight [2], making it one of the most influential organs for maintaining good health. According to a survey conducted by the World Health Organisation (WHO), amongst all diseases, skin abnormalities are the second most common worldwide, affecting 900 million people globally at any time [3]. Skin cancers (Melanoma and Non-Melanoma), warts, inflammation, bacterial and fungal infections, psoriasis, acne, rosacea, and eczema are the most commonly faced skin diseases contributing 1.79% to global disease burden [4]. In addition to that, the rate of death caused by skin cancer is three times as compared to other skin diseases [5] since it tends to spread to other body parts aggressively. Skin ailments, if not

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diagnosed and treated timely can lead to lifelong disabilities or deformations affecting the physical and psychological health of the patient.

Progression of skin ailments affects the internal layer structure; therefore, in-time diagnosis of skin disease requires analysis of skin layers. Skin is composed of three layers, i.e., epidermis, dermis, and hypodermis. Epidermis, the skin's surface layer, contains Keratinocytes, Melanocytes, Langerhans cells, and Merkel cells as components of its structure [6]. Beneath the skin's surface lies the dermis, which is further segmented into the papillary layer that contains loosely structured connective tissues interfacing with the epidermis. Dermis is succeeded by the reticular dermis, characterized by dense connective tissues and collagen fibers. Hypodermis is the innermost skin layer containing sensory nerves and blood vessels.

Various invasive and non-invasive imaging modalities [7] are widely used by histopathologists in visualizing internal skin layers and identifying disease-specific biomarkers. Brightfield microscopy, Dermoscopy, Confocal Laser Scanning Microscopy, Optical Coherence Tomography, Multi-Spectral Imaging, Multi-photon laser imaging, and 3D Topography are the most commonly known imaging techniques for skin disease diagnosis and analysis. Amongst all, brightfield microscopy is one of the most widely used imaging modalities in histopathology [8]. Brightfield microscopy requires a tissue sample extracted through biopsy, which is further preserved and processed using a chemical procedure known as staining to highlight the structural details. The resultant stained sample is further captured using a microscope at various resolutions, known as stained Whole Slide Image(WSI) sample. Amongst various staining procedures, H&E-stained tissue samples acquired from skin biopsy via electronic microscope are among the widely used modalities for skin disease detection and analysis since they highlight the internal layers and various structural anomalies through the presence of structural biomarkers.

Figure 1 demonstrates a skin cross-section indicating all layers i.e., starting from the outermost structure containing hair follicles to the innermost structure containing sensory nerves and tissues. Similar layers are highlighted in corresponding H&E-stained WSI samples depicting various structural patterns of Epidermis, Dermis, and hypodermis.

Imaging modalities discussed above require specialists to analyze acquired images, which might elapse the disease diagnosis time, thus increasing the patient's anxiety level since various benign skin cysts or anomalies mimic malignant skin cancer [11]. Furthermore, an in-time diagnosis is required to stop the disease progression and provide effective treatment. Nowadays, in addition to specialists, Computer-Aided Diagnostic (CAD) systems are employed to aid autonomous diagnosis, enhance diagnostic reliability, and reduce disease diagnosis time. Accurate segmentation of each layer is the foremost requirement in digital pathology since most skin diseases are associated with the progression

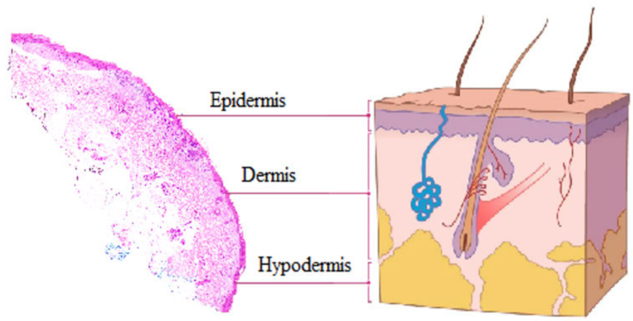

FIGURE 1. **Skin Anatomy highlighting the three important layers [10] i.e. Epidermis, Dermis, Hypodermis shown in the right portion of Figure; H&E stained biopsy sample of skin tissue is represented on the left side of Figure, where each layer has some structural features that help to differentiate them making the disease analysis more effective.**

of some structural changes in these layers. In literature, various CAD systems have been proposed to segment and extract various structures from biopsy samples, i.e., nuclei, cytoplasm, keratin, neoplasm, melanocytes, etc. However, state-of-the-art techniques still lack accurate segmentation of the region of interest, i.e., desired skin layers, to further analyze them for any disease or presence of structural evidence of anomalies. The proposed research provides a deep learning attention-based UNet framework with a transformer backbone to segment the skin layers from H&E-stained image samples accurately. Major contributions of this research work are:

- A novel U-Net-based Encoder-Decoder framework for segmentation of H&E whole slide images into Epidermis, Dermis, Hypodermis, and Keratin. The integration of a multi-axis encoder, attention mixing decoder, and attention-based skip connections has helped the model converge efficiently by reducing FLOPs and inference time compared to state-of-the-art models.
- The Multiaxis encoder, incorporating vision transformer blocks, helps to encode feature representations at various scales. Followed by the encoder is the attention-mixing decoder that has merged three attention mechanisms, including multi-head self-attention, shallow-wise convolution, and spatial and squeeze excitation attention, enabling the model to extract relevant features from all spatial dimensions. It enhances the model's ability to learn local and global contexts. Thus, in addition to capturing long-range dependencies, it helps the model capture the local context across multiple dimensions, enabling accurate segmentation of thin layers with relatively less pixel representation, such as Keratin. Adding attention-based skip connections between encoder-decoder blocks has helped the model integrate low-level and high-level features of relevant activations only while suppressing the others.
- Instead of processing the entire image and solely relying on deep features for diagnosing and classifying diseases, the study emphasizes accurate segmentation

of skin layers before disease classification. Thus, once extracted, the concerned layer can further be examined for disease-associated biomarkers.

The forthcoming paper is structured as follows: Section II discusses diverse cutting-edge approaches for skin segmentation, detection, and categorization of diseases by employing multiple imaging modalities, predominantly brightfield microscopy. Section III comprises the proposed methodology, which explains the proposed model's architectural design and discusses the dataset's specifics, followed by Section IV, i.e., the experimentation, ablation study, and results. Section V concludes the proposed research work, accentuating the research boundaries and prospects of the conducted research.

II. LITERATURE REVIEW

The literature review of conducted research has been categorized into three sub-sections based on the algorithms utilized for extracting desired information from input images, primarily WSI. The first part of the literature review consists of techniques focused on layer segmentation or extraction from WSI samples, followed by disease classification. Whereas, the second part is composed of disease classification based on deep features or structural features i.e., nuclei segmentation, contrast-based features, and textural features. In the last section of this literature review, the discussion will be centered on how several cutting-edge frameworks have been leveraging U-Net and transformers for medical image segmentation.

A. SEGMENTATION OF SKIN LAYERS OR REGIONS

In 2020 [12], a deep learning framework was proposed to detect Mycosis Fungoids, from epidermis. Since the condition originates from epidermis, to classify the disease accurately, instead of extracting deep features from the whole slide image, Epidermis was extracted from the input image using a segmentation model EU-Net extended from U-Net. The proposed model was comprised of an EfficientNet encoder trained on imageNet. The resultant segmentation map was concatenated with the input image and passed to a Convolutional Neural Network comprising convolutional blocks, max pooling followed by a fully connected layer to classify an image as diseased or healthy, yielding an accuracy of 99% on a local dataset comprising 164 WSI samples. Kleczek et. al. [13] proposed a framework to semantically segment background and foreground from H&E stained WSI images to enhance the autonomous detection of diseases from the tissue region. Since WSI contains various artifacts, background separation is a challenging task. The proposed algorithm applied a Gaussian filter on the input image to extract the noise followed by initial background segmentation using CIELab color space. The segmented image is further processed using hysteresis threshold, using a selective filtering approach that incorporates pixels included in the foreground to compute the average intensity and further refine the boundary regions. A UNet-based architecture was proposed [14] to segment input WSI patches into twelve classes that included skin layers, carcinomas, and tissues. Semantic segmentation is further proceeded by classifying input image into one of the three categories, i.e., Squamous cell Carcinoma, Basal cell carcinoma, and intra-epidermal carcinoma. The resultant framework was tested on 290 WSI samples available in three resolutions, i.e., 10x, 5x, and 2x, yielding the highest accuracy of 0.85 for 10x image resolution. Amor et. al. [15] utilized OCT skin images to segment the Epidermis and hair follicles since an accurate segmentation of the epidermis helps to analyze the presence of various skin diseases. The proposed framework consisted of a UNet segmentation model followed by Fourier domain filtering to extract the exact boundary from the input image, resulting in a dice score of 0.81. To segment epidermal from H&E-stained images, a UNet-based architecture was proposed in 2019 [16]. The proposed framework applied morphological processing on the HSV plane input image to generate ground truth, which consisted of epidermal layer contours and foreground masks. The resultant ground truth was employed to train the model for epidermis segmentation from input image. Since the input image is patched into sub-images to speed up the training process and to optimize the memory requirement, resultant patches were augmented using rotational transformations to balance the three classes, i.e., epidermis regions, other tissue regions, and background. Another framework [17] implemented threshold-based segmentation to coarsely segment the epidermis from H&E-stained images, followed by the K-Means algorithm to refine the epidermis. A dataset with 64 WSIs was used to evaluate the framework, which yielded specificity and sensitivity of 99.84% and 92.78% respectively. Osman et. al. [18] designed a framework to segment the epidermis and dermis from H&E images. The proposed framework applied thresholding and active contour method on HSV plane to extract the desired layers from the input image. Since the proposed algorithm implemented classic image processing techniques, the results may vary upon changing image contrast and input image quality. Therefore, it may not generate the desired results for images acquired via different sources. A deep learning framework [19] was proposed to extract the epidermal layer from high-frequency ultrasound images of skin using a U-shaped convolutional neural network, yielding a dice coefficient of 0.87 on a dataset of 380 images.

B. DISEASE SEGMENTATION AND CLASSIFICATION ON SKIN TISSUE IMAGES

A visually interpretable autonomous diagnostic framework [20] was proposed in 2021, which processed H&E-stained WSI samples using a deep learning framework to classify 11 types of skin diseases. The indicated research proposed a deep residual attention network using residual blocks and convolutional blocks to classify input images. To interpret the model results, attention maps were extracted to highlight the attention regions. It yielded an accuracy of 0.868 when trained and tested on 11867 WSI patches. A UNet-based architecture was proposed [21] to semantically segment Tumornest, stroma, and non-tumor region from H&E-stained WSI to detect Basal Cell Carcinoma. An ablation study was conducted using two different schemes at the decoder, i.e., deep supervision and linear. The proposed framework was tested on a local dataset with 650 WSI annotated by two derma-pathologists, yielding the highest accuracy of 96%. To accurately segment Melanoma skin cancer, a three-staged framework was proposed [22], which used H&E-stained input image, and applied K-Means algorithm to extract epidermis, followed by nuclei segmentation using a convolutional neural network. Lastly, resultant nuclei were classified as Melanoma or non-melanoma via support vector machines using HOG, Haralick textural, and Morphological features. The proposed framework yielded 0.87, 0.97, and 0.86 accuracy values for background region, active, and passive nuclei, respectively. Liu et. al. [23] proposed a deep learning framework to extract melanocytes from H&E-stained input images using masked-R-CNN pre-trained on ImageNet. Input images were pre-processed, followed by model training and testing. The resultant model yielded an accuracy of 0.927 when trained and tested on a local dataset containing 227 H&E-stained images annotated by three pathologists at 10x magnification. Another framework [24] to segment dermis, epidermis, and tumor region from H&E-stained images utilized a Fully Convolutional Network. The dataset used in the research was manually annotated into five classes, i.e., dermis, epidermis, tumor, background, and diagnostically not important. To manage the imbalance dataset, rotation and flips were utilized to augment data, resulting in an IOU score of 0.86. A semi-supervised melanocyte nests segmentation framework [25] was proposed in 2020 using auto-encoder decoder architecture with a feed-forward fully convolutional network. Firstly, ground truth was generated using a semi-supervised learning-based approach in an auto-encoder decoder to generate probability maps. Weights from the trained model were acquired to train a segmentation model to generate a binary image mask corresponding to the input image. The resultant segmentation model yielded specificity and sensitivity of 0.94 and 0.76, respectively. ConvNextV2 [26] was proposed to classify input dermoscopic images into eight classes representing skin cancer. In the first stage, the model extracted local features, followed by the second stage to extract attention-based features. The proposed framework was trained and tested on ISIC 2019 dataset resulting in an overall accuracy of 93.60%. A modified swin transformer architecture [27] with a shifting window and multi-self attention head was proposed to classify skin cancer using ISIC 2019 dataset. The proposed framework could extract fine details from dermoscopic images, resulting in an accuracy of 89.36%. A different trial [28] carried out the deployment of a MetaFormer-based model that was accurately configured to specifically classify skin cancer, which in its workflow involved the use of a focal self-attention module for help in improving and cleaning feature extraction mechanisms. The efficiency of the model was checked on ISIC 2019 and HAM10000 datasets, and the accuracy achieved was 92.54%, and 95.01% respectively.

C. TRANSFORMERS INTEGRATED WITH U-NET FOR MEDICAL IMAGE SEGMENTATION

With the emergence of various transformer architectures, medical image segmentation is becoming enhanced. In TransUNet [40], transformers have been incorporated into the U-Net encoder-decoder framework, thereby combining convolutional neural networks with self-attention mechanisms for local-global context awareness. This hybrid method achieved an average Dice score of 88.39% for segmentation tasks on MSD heptic vessel dataset. However, proposed framework may struggle to handle multimodal datasets. To tackle the apparent high computational complexity of transformers, the Vision Mamba UNet [29] architecture was devised. This model is based on a Visual State Space (VSS) block wherein the depthwise separable convolutions capture long-range dependencies efficiently while having a linear computational complexity. This framework showed promise and competitive performance with a Dice coefficient of 83.96% on the ISIC18 dataset. In addition, the Lightweight Mamba UNet [30] is another acquaintance in the field, proposed to be an efficient alternative that Mamba would act as a substitute for various CNN and Transformer blocks. LightM-UNet utilizes the Residual Vision Mamba Layer to attain deep semantic feature extraction with a low computational and parameter cost. Experimental results on 2D and 3D datasets have shown that the LightM-UNet outperforms others in the field concerning segmentation performance while being 116 times more efficient in terms of parameters and 21 times in computations. However, both VM-UNet and LightM-UNet pose a problem in dealing with complicated multimodal datasets and lack the ability to satisfactorily perform fine-grained segmentation tasks, which demand very high spatial resolution. seUNet-Trans [31] model is another transformer-based UNet architecture that optimally exploits Transformer merit within a traditional UNet framework to address the challenge of medical image segmentation. Instead of building an architecture that underutilizes the Transformer like other CNN-Transformer hybrids, seUNet-Trans uses a module that completely engages the Transformer in the segmentation pipeline. It employed pixel-wise embedding technique that eliminated the dependency on any position embedding vector during spatially reduced attention. By placing strong localization abilities of UNet combined with the strength of the Transformer in modeling long-range dependencies, the proposed framework gives a holistic understanding of medical images. The results of extensive evaluations on seven distinct medical image segmentation data sets show that the model performs better than state-of-the-art methods in terms of effectiveness. Specifically, it achieved a Dice coefficient of 0.823 in GlaS, and 0.854 in ISIC 2018 dataset.

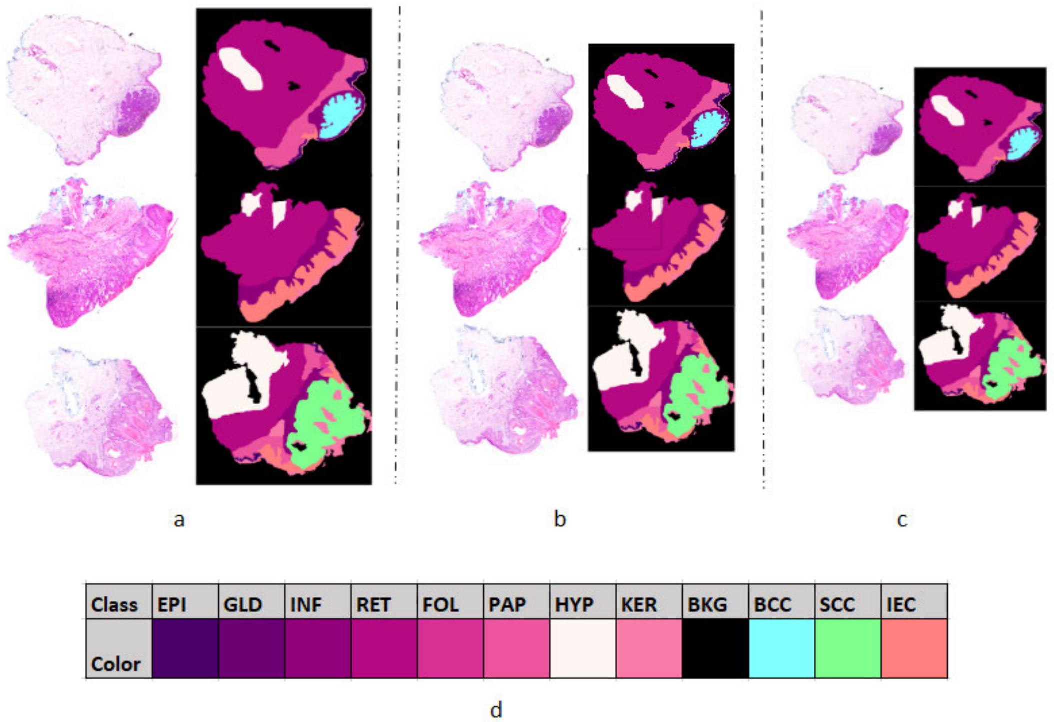

FIGURE 2. WSI samples from Queens Land Dataset; (a-c) Whole slide image paired with annotated ground truth captured at 10x, 5x, and 2x magnification level. Samples from BCC, IEC, and SCC are represented in first, second, and third row respectively; (d) Annotated images are labeled for 12 classes indicated in the color map including skin layers (EPI, PAP, HYP), skin abnormalities (GLD, INF, RET, KER), hair follicles (FOL), background (BKG) and melanoma skin cancer (BCC, SCC, IEC).

Despite the fact that seUNet-Trans performs quite well, it has limitations when it comes to computational efficiency with high-quality images, as well as the model faces challenges when dealing with multimodal datasets that require divergent representation of features. DA-TransUNet [32] another modification to transformer architecture has synthesized two attention mechanisms. It was able to improve segmentation performance because the integrated systems could utilize local and global contextual inputs. The reduced noise allows focusing on more relevant features, while less relevant ones get filtered out, hence making the segmentation process very fine with a dice score of 79.80% on Synapse dataset. However, an increased computational overhead makes the entire framework rather unfit for environments with limited resource availability.

The literature review concluded that the majority of frameworks proposed for disease classification of skin, instead of focusing on extracting the desired layers, utilized raw images to extract deep features or contrast-based and textural features for classifying an image using machine learning algorithms. Therefore, such models lack interpretability since one cannot identify the reasons for the misclassification of input image and, therefore cannot be reliable in the case of CAD systems which require a high degree of precision and recall. Moreover, since each disease targets a certain layer, extracting features from the concerned region will add high precision. Secondly, most layer segmentation algorithms focus solely on epidermis segmentation, followed by disease classification based on the extracted layer. However, many other skin diseases influence the structure of other skin layers, i.e., dermis, hypodermis, and keratin; therefore, accurate segmentation of all layers is crucial for disease diagnosis. To improve segmentation accuracy, transformer-based encoder-decoder architecture has been widely used in medical image segmentation. However, transformer-based configurations in medical image segmentation have to contend with the trade-off between computational complexity and feature representation. Although transformers are best at capturing long-range dependencies and global context, their exorbitant computations become a hindrance when reflecting on real-time implementations of techniques, especially in constrained environments. Optimization attempts, such as spatially reduced attention or hybrid CNN-Transformer frameworks, hinder precise segmentation. Thus, this trade-off between computational efficiency and feature richness makes Transformer-based medical image segmentation a cause for concern regarding its further advancement. This study proposes a transformer-based UNet model for segmenting skin layers significant in disease diagnosis, such as the epidermis, dermis, hypodermis, and keratin. In the proposed research, the multi-axis encoder is followed by the attention-mixing decoder to enhance feature extraction from multimodal data for higher precision in skin layer delineation with distinct boundaries. Additionally, attention-based skip connections allow the model to consider relevant regions only, making the model computationally less expensive without affecting the performance. Accurate segmentation of these layers, therefore, contributes to a more rapid pipeline in skin diagnosis, enabling layer-specific identification of disease-associated biomarkers for further studies.

III. MATERIALS AND METHODS

A. DATASET

Conducted research utilized a publicly available dataset, i.e., Queensland [33], comprised of 290 H&E stained images. The images are classified into three groups: Basal Cell Carcinoma with 140 images, Squamous Cell Carcinoma with 60 images, and Intra-Epidermal Carcinoma with 90 images. When categorized by biopsy types, this dataset is composed of 100 shave, 58 punch, and 132 excisional biopsies. Each tissue is paired with its ground truth annotated for 12 labels indicating various tissue regions (glands, inflammation, hair follicles), cancer regions (Squamous Cell Carcinoma, Basal Cell Carcinoma, Intra-Epidermal carcinoma), and background. In the abovementioned dataset, whole slide image samples were prepared by acquiring tissue samples from patients with an age range between 36 and 90, with a median age of 70, comprising two-thirds of females and one-third of males. Since input samples were captured using a high-resolution Olympus brightfield microscope, the resultant images have a high resolution ranging between 11 million to 500 million pixels. Each tissue slide was captured at three magnification levels from high to low, i.e., 10x, 5x, and 2x. Figure 2 contains a sample original image and its associated ground truth from all three categories of carcinomas, where each pixel has been assigned a class color as represented in the color map.

B. METHODOLOGY

The proposed methodology is primarily of sub-stages, i.e., pre-processing input images, model training, and validation followed by testing the trained model. WSI samples available in the dataset are of high resolution, which requires high memory and processing time while loading the entire dataset along with ground truth. Therefore, input image samples are pre-processed and patched before passing them to the training pipeline. Additionally, accurate segmentation of skin layers is crucial for localizing skin diseases [34]. Therefore, the proposed framework is trained to segment five medically important layers, namely the Epidermis, Hypodermis, Dermis (Reticular and Papillary), and Keratin [1], from whole slide images. Figure 3 elaborates the three phases of the proposed pipeline, i.e., pre-processing, model training, and inference.

1) PRE-PROCESSING

To reduce memory consumption, input images are further patched into 256×256 non-overlapping patches. Additionally, to generalize model training, input images are augmented using transformations such as flip, rotate, and invert by varying 50%-180% from the original position. Apart from the input image preprocessing, the corresponding ground truth was also preprocessed. Since the ground truth is annotated for 12 layers dedicated to different tissue structures and cancer regions; however, this research focuses on the extraction of dermis, epidermis, hypodermis, and keratin. Therefore, ground truth was modified by assigning 0-5 labels to pixels belonging to background, hypodermis, reticular dermis, papillary dermis, epidermis, and keratin. The rest of the tissue regions were labeled '0' and treated as the background class.

C. MODEL ARCHITECTURE

Medical image segmentation can achieve phenomenally high accuracies using transformers owing to their self-attention mechanism, which caters well to long-range dependencies. While classic transformer models have their benefits, they generally struggle to extract local features in combination with global features for multimodal medical data. A viable and powerful linear self-attention framework is computationally expensive; thus, it becomes a serious challenge in poorly resourced environments.

To overcome these limitations, a novel encoder-decoder-based architecture for medical image segmentation is proposed. The encoder utilizes a pre-trained multi-axis vision transformer, which is capable of disentangling the global context and local context with respect to each region of the medical image. The multi-axis approach combines axial and spatial attention, enabling the model to process complex tissue structures with varying spatial relationships.

The decoder wonderfully stitches feature maps across scales while retaining spatial coherence using the Convolutional Attention Mixing module. The convolutional attention mixing module mixes multi-head self-attention with shallow-wise convolution and spatial and squeeze excitation: this permits efficient modeling of long-range dependencies while lowering computational cost. The shallow-wise convolution helps local feature extraction; the spatial and squeeze excitation modules help recalibrate their responses, thus amplifying the quality of segmentation. Attention-based skip connections in the decoder selectively transfer spatial and contextual features from the encoder. These connections apply attention weighting to the created features, allowing the model to focus on the most informative regions for better segmentation performance.

In Figure 4, the proposed architecture is illustrated, elaborating the decoder modules and the detailed flow of information through the attention-based skip connections. This design guarantees that the model is adequately tuned to capture critical fine details with sufficient consideration

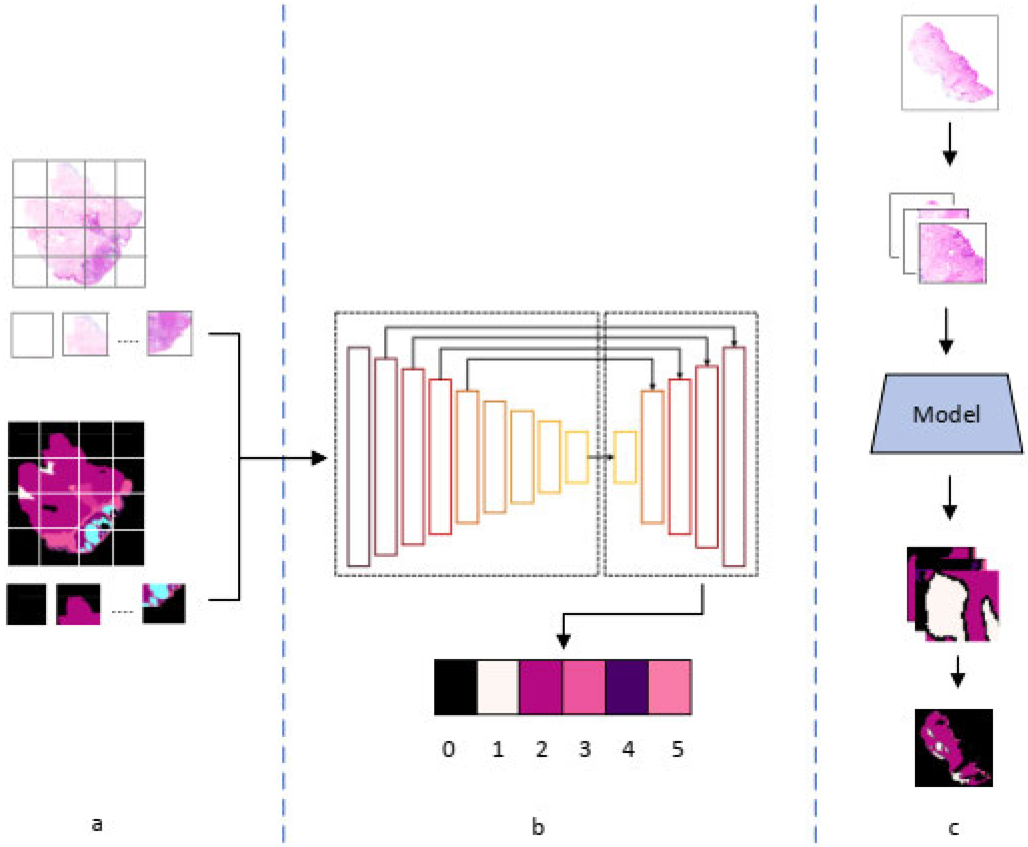

FIGURE 3. Proposed framework for the layer segmentation using whole slide images acquired using biopsy; (a) Pre-processing of WSI samples i.e., ground truth preparation and patching; (b) Model Training on pre-processed patches for five classes i.e., BKG, HYP, RET, PAP, EPI, KER as indicated in color map respectively; (c) Inference pipeline for the trained model which takes patched WSI samples and segment them. Resultant patches are merged to form a single WSI sample.

of the broader context, vital for segmentation success in dermatopathology. The framework achieves a fair balance between computation demands and segmentation accuracy by combining these architectural elements, thus readily rendering it to real-world clinical applications where both accuracy and efficiency are critical.

1) MULTI-AXIS ENCODER BACKBONE

In the encoder block, a pre-trained multi-axis vision transformer (MaxViT) is utilized due to its proven effectiveness in the medical imaging field. MaxViT encoder blocks are employed to extract features from the input image at each resolution. The encoder processes an RGB input image with dimensions 3 × 256 × 256, passing it through initial stem blocks with sizes of 32 and 64. The encoder block contains four sub-blocks, where each sub-block is further comprised of pre-trained MaxViT blocks. To incorporate local and global feature extraction at each level, each MaxViT block contains three elements, i.e., MBConv block, axial block attention, and grid attention. MBConv is responsible for the proficient acquisition of local characteristics and intricate details. The implementation of depth-wise separable convolutions, an inverted bottleneck architecture, and squeeze-and-excitation mechanisms enables it to concentrate on accurate and localized feature extraction. MBConv enhances computational efficacy while proficiently preserving spatial information, thus establishing it as an essential component for the acquisition of local dependencies in the foundational layers. MBConv block is mathematically represented as:

$$X_{expand} = Conv_{1\times 1}(X) \in \mathbf{R}^{H\times W\times (t.C)} \tag{1}$$

$$X_{depthwise} = DepthwiseConv_{k\times k}(X_{expand}) \in \mathbf{R}^{H\times W\times (t.C)} \tag{2}$$

where X is the input feature map, and H, and W are the height and width of feature maps. The number of channels is represented as C. For channel recalibration, apply a squeeze operation by global average pooling followed by excitation (scaling the channels):

$$z = GlobalAvgPool(X_{depthwise}) \in \mathbf{R}^{t.C} \tag{3}$$

$$X_{scale} = X_{depthwise}.s(z) \in \mathbf{R}^{H\times W\times (t.C)} \tag{4}$$

$$X_{MBConv} = Conv_{1\times 1}(X_{scale}) \in \mathbf{R}^{H\times W\times C} \tag{5}$$

In the block attention module, the feature map is divided into non-overlapping regions thus enabling attention to be

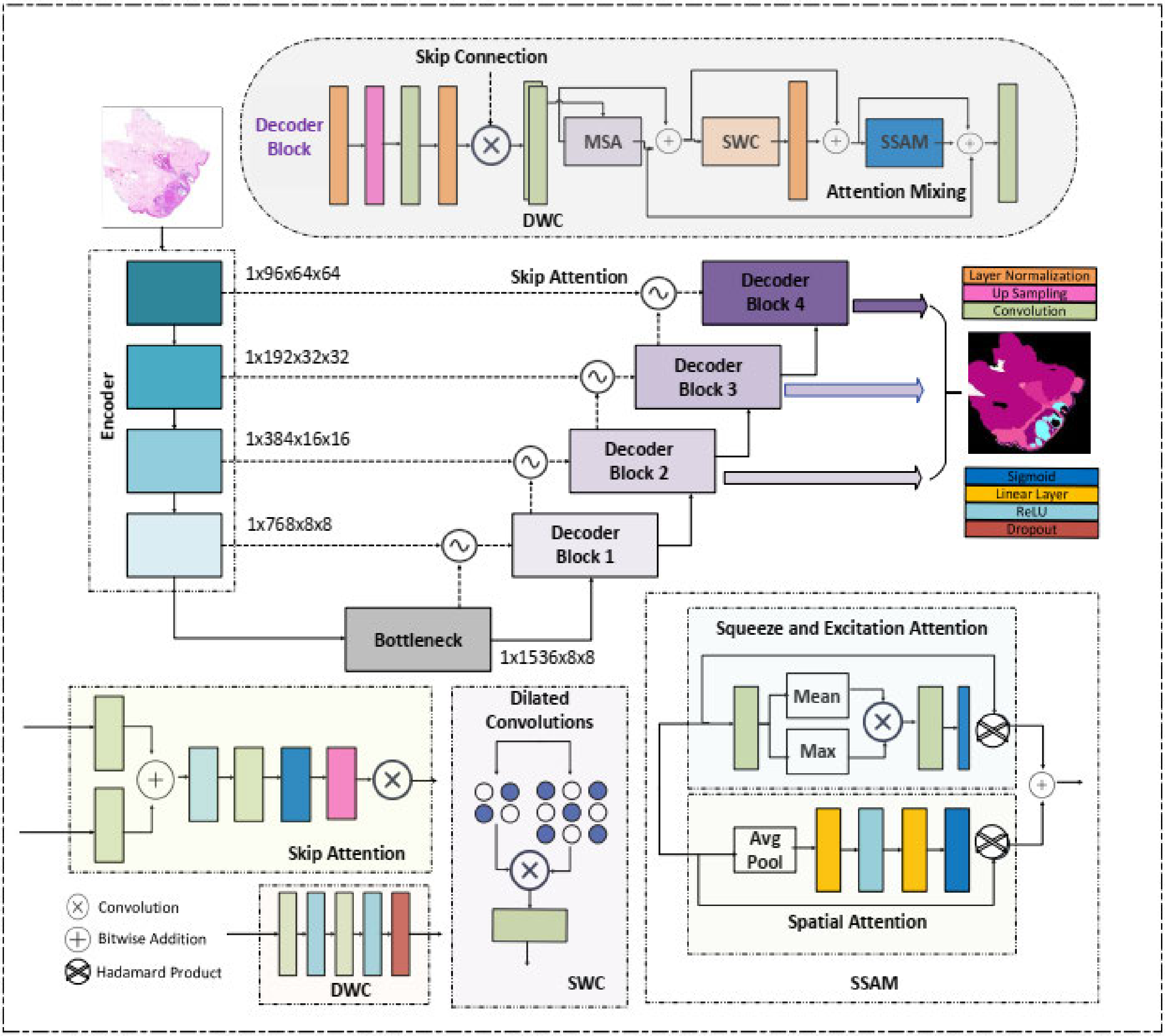

FIGURE 4. **The detailed architecture of the proposed framework is composed of ViT-based pre-trained encoder blocks connected to their corresponding decoder block via attention-based skip connection. The multiaxis encoder helps the model to learn global and local contextual details, which are then passed to the decoder block. Attention gates at skip connections allow the passing of attention at relevant regions while suppressing the rest. Attention mixing at the decoder end helps the model retain long-range dependencies and merge the features extracted at various resolutions.**

applied within each block independently. Output from the previous block i.e., X_{MBConv} is divided into $\frac{H \times W}{B^2}$, where each block $B_i \in \mathbf{R}^{\mathbf{B} \times \mathbf{B} \times \mathbf{C}}$. Key, Query, and value matrices are computed as:

$$Q_i = W_q B_i \tag{6}$$

$$K_i = W_k B_i \tag{7}$$

$$V_i = W_v B_i \tag{8}$$

Attention within the blocks is represented as:

$$Attention(Q_i, K_i, V_i) = softmax(\frac{Q_i K_i^T}{\sqrt{d_k}})V_i \tag{9}$$

where d_k is the dimension of query and key. The final feature map is formed by concatenating all blocks, mathematically represented as:

$$Y_{BlockAttn} = concat(B_1, B_2, B_{\frac{H \times W}{B^2}}) \tag{10}$$

In the grid attention mechanism, the attention mechanism is applied to the entire grid, thus enabling the model to learn the global context. It correlates the information learned in different blocks by focusing on spatial relationships throughout the grid. This global attention mechanism enables MaxViT to understand broader patterns and global dependencies, which is particularly beneficial for complex patterns and shapes covering larger image areas. Feature map acquired from block attention module is represented as $Y_{BlockAttn} \in \mathbf{R}^{\mathbf{H}\times\mathbf{W}\times\mathbf{C}}$, which is split into grid cells with dimension $G \times G$. Key, Query, and Value matrices for each grid cell are represented as:

$$Q_j = W_q G_j \tag{11}$$
$$K_j = W_k G_j \tag{12}$$
$$V_j = W_v G_j \tag{13}$$

Apply attention across the grid cells, allowing information exchange across the global grid:

$$Attention(Q_j, K_j, V_j) = softmax(\frac{Q_j K_j^T}{\sqrt{d_k}})V_j \tag{14}$$

To generate the final feature map, grid cells are concatenated as:

$$Y_{GridAttn} = concat(G_1, G_2, G_{\frac{H\times W}{G^2}}) \tag{15}$$

After traversing through MBConv, block attention, and grid attention, the final output feature map $Y_{GridAttn}$ furnishes a multi-scale representation that combines local features, mid-range context, and global dependencies, enabling MaxViT to execute proficiently on visual tasks that necessitate an equilibrium of detailed and contextual comprehension. To incorporate the vanishing gradient problem, each block is equipped with skip connections. MaxViT backbone generates four feature maps represented as $1 \times (96 \times 2^i) \times \frac{H}{2^{2+i}} \times \frac{W}{2^{2+i}}$, where $i = 0, 1, 2, 3$ represent stages.

2) CONVOLUTIONAL ATTENTION MIXING DECODER

The decoder reconstructs the segmented output from the compressed features by upsampling (using interpolation) and gradually refining the spatial details. There are four decoder blocks in correspondence with four encoder stages, working from low to high resolution to produce the final segmentation map. Each decoder initiates with layer normalization, followed by up-sampling to match the size with skip connections from the encoder output. 2D convolution is applied to the resultant matrix, followed by layer normalization. Finally, the results are concatenated with the output from the gated attention block through skip connection, mathematically represented as:

$$X = Concat(N(C_{2D}(U(N(X)))), GA(Enc_{(i)}, Dec_{(i-1)})) \tag{16}$$

where N illustrates Normalization, C_{2D} represents 2D convolution, U represents up-sampling, and X indicates the input to the decoder block. Instead of plain skip connections, a gated attention block has been employed to help the model learn from the attention of relevant regions. Gated attention block is mathematically represented as:

$$\begin{aligned} GA(Enc_{(i)}, Dec_{(i-1)}) \\ = \alpha(Enc_{(i)}, Dec_{(i-1)}) \times Enc_{(i)} \end{aligned} \tag{17}$$

$$\alpha(X, g) = Upsample(\sigma_2(\psi(\sigma_1(W_g(g) \bigoplus W_x(X)))), 2) \tag{18}$$

Here $Enc_{(i)}$ represents the output from the encoder block at the i^{th} level, $Dec_{(i-1)}$ indicates the gated output from the decoder block at the previous level, where $i = 1, 2, 3, 4$. In the proposed architecture, instead of directly propagating the information from the encoder block to the decoder block, the attention-based mechanism adds weight to the passing parameters. W indicates a 1×1 convolution with a stride of 1 and 2 for W_g and W_x respectively. To retain the output range, ReLU (σ_1) and Sigmoid (σ_2) functions are applied, coupled with

$$1 \times 1$$

convolution (ψ) with a filter size of 1 to reduce the dimensions of the final weight matrix. The resultant weight matrix is upsampled by a factor of 2 to match the size with output from the current encoder block i.e., $Enc_{(i)}$.

The concatenated output is further passed through depth-wise convolution which helps extract the relevant features and reduce the channel dimension by a scale of two. Depth-wise convolution is mathematically represented as equation 19. Where D, R, and C_i represents dropout, ReLU and 2D Convolution and $i = 1, 2$.

$$DWC = D(R(C_2(R(C_1)))) \tag{19}$$

DWC when applied to X is represented as:

$$X' = DWC(X) \tag{20}$$

Followed by depth-wise convolution, multi-self attention is applied, where linear projections are replaced by convolutional projections to reduce computational complexity. Utilizing convolutional projections has also helped in acquiring more spatial context and reduces the number of parameters. The number of attention heads in each decoder block is $N = 10, 8, 6, 4, 2$ respectively. Input and output of Multi-head self-attention block are added through skip connections, mathematically represented as:

$$X'' = MSA(q, v, k) + X'; q, v, k = N(R(C_j(X'))) \tag{21}$$

where N and C_j indicate the normalization and convolution layers respectively for query, key, and value represented as $j = 1, 2, 3$. Added feature maps are then passed through shallow-wise convolution that facilitates capturing local and global image details at each level. Input and output feature maps from shallow-wise convolution are concatenated after applying layer normalization on the output feature mathematically represented as:

$$SWC = C_1(Concat(X_1'', X_2'')) \tag{22}$$
$$X_1'' = R(C_2(X'', d = 2)), X_2'' = R(C_3(X'', d = 3)) \tag{23}$$
$$X''' = N(SWC(X'')) + X'' \tag{24}$$

where $Concat$ indicates concatenation, d indicates the dilation term applied during convolution to expand the kernel width. The resultant feature map is then passed through the spatial attention and squeeze excitation attention modules (SSAM).

$$SSAM = SA(X''') \times X''' + SEA(X''') \times X''' \quad (25)$$

where

$$SEA = \sigma_1(C_3(Concat(Mean(C_2(X''')), Max(C_1(X'''))))) \quad (26)$$

$$SA = \sigma_2(L(R(L(P_a(X'''))))) \quad (27)$$

where σ_i, L, and P_a indicate sigmoid, linear layer, and adaptive average pooling layer. To get the final feature map, feature output from multi-head self-attention, Spatial, squeeze, and excitation attention modules are added, mathematically represented as:

$$X'''' = SSAM(X''' + X''' + X'') \quad (28)$$

In the discussed architecture, a kernel of size 3×3 is utilized in convolution. The implementation requires an RGB image with size $3 \times 256 \times 256$. Output feature map from decoder results in four matrices with size $(96 \times 2^i) \times \frac{H}{2^{(2+i)}} \times \frac{W}{2^{(2+i)}}$ where $i = 0, 1, 2, 3$. Resultant feature maps are then passed through convolutional layers to reduce the dimension to the number of classes N_{cls}.

The encoder-decoder architecture requires an RGB image, to generate four feature maps (E_1, E_2, E_3, E_4), where the last feature map is passed through the bottleneck layer. On the decoder side, each feature map has increased in size by a factor of two to match the outputs from the encoder through skip connections. The final segmentation map is extracted by superposition of feature maps from the second, third, and fourth decoder block (D_2, D_3, D_4) represented mathematically as:

$$Y_{Ncls} = D_2 + D_3 + D_4 \quad (29)$$

The proposed architecture has been employed for semantic segmentation on our input whole slide image, categorizing it into one of the classes ranging from 0 to 5. In contrast to other segmentation models like UNet and Fully Convolutional Network, the utilized architecture has demonstrated remarkable progress in segmentation accuracy [36], [37], [38]. Additionally, the attention-based skip connection has helped the model converge faster as compared to the regular UNet-based architecture.

3) POST-PROCESSING

Once the image is inferred from the trained model, the resultant patches are labeled for classes (0-(5) pixel-wise. Resultant labeled images are then processed to assign colors based on class values to generate colored labeled images using the color dictionary highlighted in Figure 2. The resultant patched predictions are then merged to generate a single WSI sample indicating all the layers and background regions. To further smoothen the generated images and ensure continuity amongst the resultant image, a closing filter with a kernel size of 5×5 is applied, followed by a Gaussian filter. Figure 5 moving from left to right demonstrates the original image from the dataset, associated ground truth containing labels for classes (0-(11), pre-processed image with updated class labels highlighting skin layers only (0-(5), and the inferred images from the trained model. It is evident from the segmented image when compared with ground truth, that the model has accurately segmented the desired layers, making the disease diagnosis procedure effective and accurate.

IV. EXPERIMENTATION AND RESULTS

The proposed framework has been implemented using PyTorch (version 1.11.0 with CUDA 11.3 support) on NVIDIA GeForce RTX 2070 with 8GB RAM. An extensive study on the evaluations made on major loss functions applicable to medical image segmentation showed that the loss aggregation approach, by mashing features from different stages together, would ensure improvement in model convergence as well as an increase in its efficiency for performing image segmentation tasks. Under the provided loss function, the neural network produces n prediction maps. In computing the loss, all $2^{(n-1)}$ non-empty subsets of these prediction maps are taken into account. Each subset is aggregated to create one consolidated prediction map, which is then used for the loss function evaluation at the end. This feature mixing methodology produces $2^{(n-1)}$ distinct prediction maps, encompassing the original prediction maps produced by the model. Subsequently, DICE (L_{DICE}) and Cross-Entropy (L_{CE}) loss functions are applied across all 2^{n-1} individual prediction maps alongside their respective segmentation masks to assess the losses. All computed losses are aggregated using the cross-entropy loss function, as represented in Equation 30, to derive the final loss mathematically expressed in Equation 31. The model has been trained using a batch size of 8, with a learning rate of 6×10^{-6} and Adam Optimizer for 50 epochs, and a combination of loss functions as described in the equation has been used to converge the model.

$$L_{CE} = -\sum_{c=1}^{M} y_{o,c} \log(P_{o,c}) \quad (30)$$

$$L_{Total} = \gamma L_{DICE} + (1-\gamma) L_{CE} \quad (31)$$

where $\gamma = 0.3$ and $(1-\gamma) = 0.7$ are the weights for L_{DICE} and L_{CE} respectively. The conducted research has been evaluated using a publicly available dataset, comprising tissue samples from the skin. The resultant framework has been evaluated both quantitatively and qualitatively using a test train split of 80% and 20%, where various evaluation metrics are utilized for quantitative evaluation, and class activation maps are utilized for enhancing model interpretability. The upcoming section discusses the model evaluation results in detail.

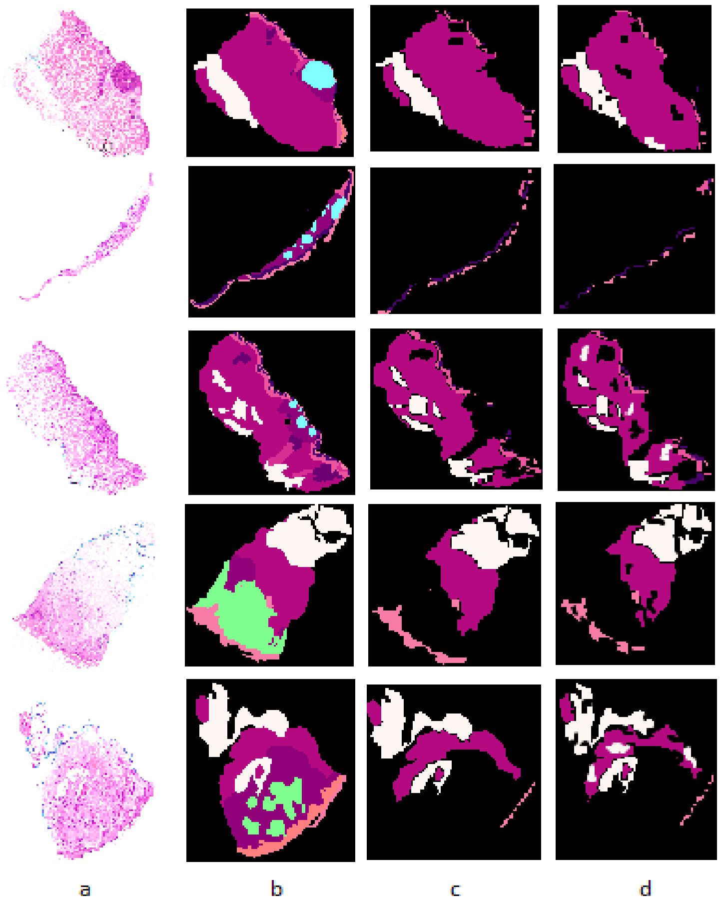

FIGURE 5. Image Segmentation Results for the proposed model trained and tested on 10x resolution; (a) WSI samples captured using bright-field microscope;(b) Ground truth with 12 classes including glands, hair follicles, layers and cancer (0-(12); (c) Ground truth representing skin layers and background i.e., BKG, HYP, RET, PAP, EPI, KER; (d) Output images inferred from model and post-processed to form a single WSI.

A. EVALUATION METRICS

To evaluate the proposed framework, Accuracy, F1-Score, Recall, DICE score and Hausdorff Distance (HD) have been used as represented in the following mathematical equations.

$$Recall = \frac{TP}{TP + FN} \tag{32}$$

$$Precision = \frac{TP}{TP + FP} \tag{33}$$

$$F1 - Score = \frac{TP}{TP + \frac{1}{2}(FP + FN)} \tag{34}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{35}$$

$$DICE(X, Y) = \frac{2 \times |X \cap Y|}{|X| + |Y|} \tag{36}$$

$$\begin{aligned} HD(X, Y) &= \max\{d_{XY}, d_{YX}\} \\ &= \max\left\{\max_{x\in X}\min_{y\in Y} d(x, y), \max_{y\in Y}\min_{x\in X} d(x, y)\right\} \end{aligned} \tag{37}$$

where True Positives, False Positives, True Negatives, and False Negatives are represented as TP, FP, TN, and FN, respectively. Moreover, X represents the ground truth, and Y represents the segmented image.

B. ABLATION STUDY

A series of experiments was designed to achieve a thorough analysis of the capabilities of the proposed architecture. The objective of the first experiment was to compare the segmentation accuracy of the proposed framework with competing state-of-the-art models, therefore establishing relative performance. The second experiment followed, consisting of a parametric study that analyzed the effects of modifying parameters in the decoder module and went even further in assessing the effectiveness of the model with a comparison against the baseline model. The experiment would then assess segmentation accuracy in key skin layers: Epidermis, Dermis, Hypodermis, and Keratin. Furthermore, the architecture was put to the test by training it with all 12 annotated classes in the dataset for comparison against the baseline model.

TABLE 1. Performance comparison of various state-of-the-art segmentation models trained on Queensland dataset with respect to various evaluation parameters, i.e., overall accuracy, F1-Score, and DICE. Stated models have been trained on all resolutions, i.e., 10x, 5x, 2x, separately.

Models	Accuracy (%)			F1-Score (%)			DICE (%)		
	10x	5x	2x	10x	5x	2x	10x	5x	2x
SegFormer [35]	89.1	87.3	86.0	88.2	85.8	84.5	87.7	89.2	88.7
TransUNet [40]	91.7	89.6	89.5	90.1	88.6	87.1	90.9	89.4	87.1
SwinUNet [41]	92.4	90.6	90.2	91.8	89.3	87.7	90.5	89.3	87.1
MT-UNet [42]	92.1	89.7	89.2	90.1	89.8	88.4	91.1	90.7	87.8
MISSFormer [38]	92.5	90.1	91.1	89.7	90.2	89.8	92.0	91.3	89.3
PVT-CASCADE [44]	92.7	91.4	91.8	90.5	90.0	90.3	92.8	91.7	89.9
nnUNet [45]	92.9	92.1	91.9	91.3	90.8	89.1	92.5	91.9	89.5
TransCASCADE [43]	93.0	90.5	91.5	92.5	91.7	90.9	92.9	90.5	89.2
nnFormer [46]	93.8	91.7	91.9	92.3	92.8	91.1	92.1	90.5	90.2
Parallel MERIT [47]	93.1	91.5	90.7	93.2	92.5	91.5	92.5	90.7	90.5
Proposed Model	**95.5**	**93.7**	**91.5**	**94.0**	**92.7**	**91.6**	**94.6**	**93.3**	**91.2**

Lastly, to test the robustness of the proposed framework, another publicly available dataset was utilized to evaluate its performance.

1) COMPARISON WITH STATE-OF-THE-ART SEGMENTATION MODELS

To evaluate the performance of the utilized architecture, various state-of-the-art segmentation models featuring transformer backbones and U-shaped architectures were compared. The models included in the comparison are SegFormer, TransUNet [40], SwinUNet [41], MT-UNet [42], MISSFormer [38], PVT-CASCADE [43], nnUNET [44], TransCASCADE [45], nnFormer [46], and Parallel MERIT [47]. Table 1 indicates a detailed comparison of the trained model with state-of-the-art segmentation models widely being used for medical image segmentation.

As evident from the results, our model performed best compared to the rest of the models. When evaluated by shuffling the train test images, the model resulted in a mean accuracy of 95.57 $\pm$ 0.06, mean F1-Score of 94.06 $\pm$ 0.08, and mean DICE score of 94.67 $\pm$ 0.05 for ten different experiments. As illustrated in the Table 1, 2% improvement has been observed when compared with the Parallel MERIT model, whose encoder architecture is similar to the MaxViT encoder, however in the decoder block shallow-wise convolution, Multi-head self-attention with convolutional projection, and Spatial excitation module results in capturing both local and global context from all spatial dimensions. Furthermore, the convolutional multi-head self-attention (MSA) facilitates the acquisition of long-range dependencies for each pixel, alongside a more comprehensive spatial pixel representation in contrast to the linear projected MSA. Additionally, shallow convolutions (SWC) were employed, utilizing various dilated convolutions to extract salient information and expand the kernel's receptive field, potentially enhancing the overall performance of the model. The attention-based skip connections have also enabled the model to train based on spatial and contextual features from relevant regions. Moreover, it's evident from the detailed experimentation that the most accurate segmentation results have been achieved for dataset acquired at 10$\times$ magnification. Although 10$\times$ magnification level reduces the field of view, i.e., whole tissue might not be visible on a single slide, however texture and structural details are clearer, making the model capture more texture-based details and helping the model learn better.

TABLE 2. The class-wise segmentation performance comparison of the state-of-the-art models on the Queensland dataset captured at 10x. The model shows better accuracy in all layers of the skin, especially in the tougher regions of the epidermis and keratin layers.

Models	BKG	HYP	RET	PAP	EPI	KER
SegFormer [35]	97.3	94.2	87.7	88.3	82.8	84.1
TransUNet [40]	97.7	95.1	91.5	90.6	86.8	87.4
SwinUNet [41]	98.1	94.5	92.6	92.9	87.1	89.7
MT-UNet [42]	98	93.8	91.1	91.6	89.8	88.7
MISSFormer [38]	98.8	94.1	93.3	92.7	87.3	88.5
PVT-CASCADE [44]	97	94.6	92.9	94.5	87.9	89.1
nnUNet [45]	98.4	93.4	95.9	96.1	85.7	87.8
TransCASCADE [43]	97.1	94.8	96.3	95.5	85.3	86.7
nnFormer [46]	98.3	95.4	97.2	96.6	83.2	86.9
Parallel MERIT [47]	98.7	96.1	97.9	95.9	82.1	85.5
Proposed Model	**98.4**	**97.4**	**96.9**	**97.5**	**91.5**	**91.8**

The model was further evaluated for each class using whole slide images captured at 10x magnification, as illustrated in Table 2. The model outperforms the existing methods in segmentation for the Hypodermis (97.4%), Reticular Dermis (96.9%), and Papillary Dermis (97.5%), making it robust in differentiating from even deeper skin structures. Besides, it has the highest accuracy for the Epidermis (91.5%) and Keratin (91.8%), which increases significantly compared with other models, like Parallel MERIT (82.1% EPI, 85.5% KER) and nnFormer (83.2% EPI, 86.9% KER) which raised barriers in segmentation on this class, demonstrating the efficiency of the proposed methodology in modeling thin, irregular, morphologically complex layers that frequently prove problematic in the segmentation task. Additionally, for the Background (98.4%), the proposed model performs almost at par with leading models like MISSFormer (98.8%)

TABLE 3. Analysing computational efficiency of the proposed framework when compared with state-of-the-art models. The attention-based decoder has helped the model extract relevant features and has reduced FLOPs and inference time effectively.

Models	Params (M)	FLOPs (G)	Inference Time (ms)	Accuracy (%)
SegFormer [35]	81.4	25.6	28.9	89.1
TransUNet [40]	105.3	45.2	48.65	91.7
SwinUNet [41]	27.2	37.2	34.82	92.4
MT-UNet [42]	61.4	23.8	25.12	92.1
MISSFormer [38]	92.5	37.8	42.86	92.5
PVT-CASCADE [44]	94.1	30.5	27.94	92.7
nnUNet [45]	109.8	55.4	62.76	92.9
TransCASCADE [43]	101.3	32.7	35.28	93.0
nnFormer [46]	142.8	73.0	81.23	93.8
Parallel MERIT [47]	147.5	39.3	45.11	93.1
Proposed Model	**95.74**	**28.7**	**32.76**	**95.5**

and Parallel MERIT (98.7%) to ensure the non-tissue regions are identified properly. These findings imply that combining the attention mechanism in the decoder has enhanced feature extraction and class-specific refinement. Further, it has advanced contextual learning by extracting features from all spatial dimensions, making the model learn intricate details. Additionally, gated skip connections have helped the model eliminate the irrelevant regions, and focus on the regions with more contextual relevance.

The comparison of the computational aspect of the proposed framework to state-of-the-art segmentation models is shown in Table 3. The proposed model achieves higher segmentation accuracy at lower computational cost, thus outpacing existing methods. The attention-based architecture in the decoder block allows effective feature extraction within an optimal region of trade-off regarding computational complexity and performance.

The proposed model is comprised of 95.74 million parameters compared to those of other transformer models like TransUNet (105.3 million), MISSFormer (92.5 million), and nnFormer (142.8 million). For its size, it is quite light in terms of computation and requires only 28.7 GFLOPs, which greatly lowers the computation burden in comparison to nnFormer (73.0 GFLOPs) and TransUNet (45.2 GFLOPs). Besides, inference time becomes important, especially in the case of large-scale or real-time medical image segmentation applications. The proposed model attains an inference time of 32.76 milliseconds/image, significantly exceeding the performances of heavier architectures, such as nnFormer (81.23 ms), nnUNet (62.76 ms), and TransUNet (48.65 ms). Although some networks like MT-UNet have slightly faster inference results (25.12 ms), they account for low scores for segmentation accuracy compared to our method.

To statistically analyze the proposed framework and assess its robustness, a p-value test was conducted. The p-values were calculated using paired statistical tests to compare the DICE scores obtained from the proposed model with those of the different state-of-the-art models across all test images. Figure 6 shows the distribution of p-values of different models across all test images ranging from 0 to 0.03, whereby the lower values represent strong enhancement in statistical significance, thereby emphasizing that the observed improvements in segmentation accuracy cannot be attributed to random chance. From the experiments performed, the proposed model stood out significantly by consistently attaining lowest p-values, endorsing the statistical robustness of the alleged improvements in performance. The large difference from existing models indicates that our model generalizes better across different test sets while maintaining superior segmentation accuracy when shuffling train-test images. The divergence found in the p-values across models further signifies that while some architectures exhibit a modest improvement, indeed, our framework exhibits apparent statistical improvements owing to its hybrid attention-convolutional design. The box plot shows that the proposed model also has lower variance in terms of p-values, demonstrating steady improvement across various test images.

2) TWEAKING DECODER BLOCK

To further analyze the performance of the proposed architecture, various experiments were conducted by varying the calibrations of decoder block. The model was further evaluated by adjusting the attention mixing, modifying the projection type of the multi-head self-attention module, and altering the dilations in the shallow-wise convolution. For a detailed analysis, various combinations were explored, including attention mixing (yes, no), Multi-Head Self-Attention projection (linear, convolution), dilation combinations ({2,3}, {1,2}), and attention accumulation methods (concatenation, summation). Against all the empirical analysis, the best results were achieved by retaining the attention mixing, utilizing convolutional projection, setting dilation to 2,3, and concatenating the results to amalgamate the attention module results. To illustrate the impact of each parameter on the model's accuracy, the values of all parameters were kept constant while varying one at a time, as shown in Table 4. Adding attention mixing in the decoder block has resulted in increased accuracy by 7% as compared to the model without attention mixing, since it helps the model capture both global and local context.

Using convolutional projection has helped the model learn by reducing the computational complexity i.e., the parameters have been reduced from 107.23 M to 95.74 M, thus helping the model converge in relatively less time. Similarly, more dilation by a large factor helps the model acquire deep features and thus learn more contextual and textural-based features resulting in an improvement of 7%. Lastly, concatenating the feature blocks in the attention mixing module of the decoder has improved the performance of the model by 3%, since it helps retain all the extracted features.

Another ablation study evaluated different combinations of multi-head self-attention (MSA), shallow-wise convolution (SWC), spatial and squeeze-excitation attention (SSAM), and attention-based skip connections to further analyze the impact of various attention mechanisms on the proposed

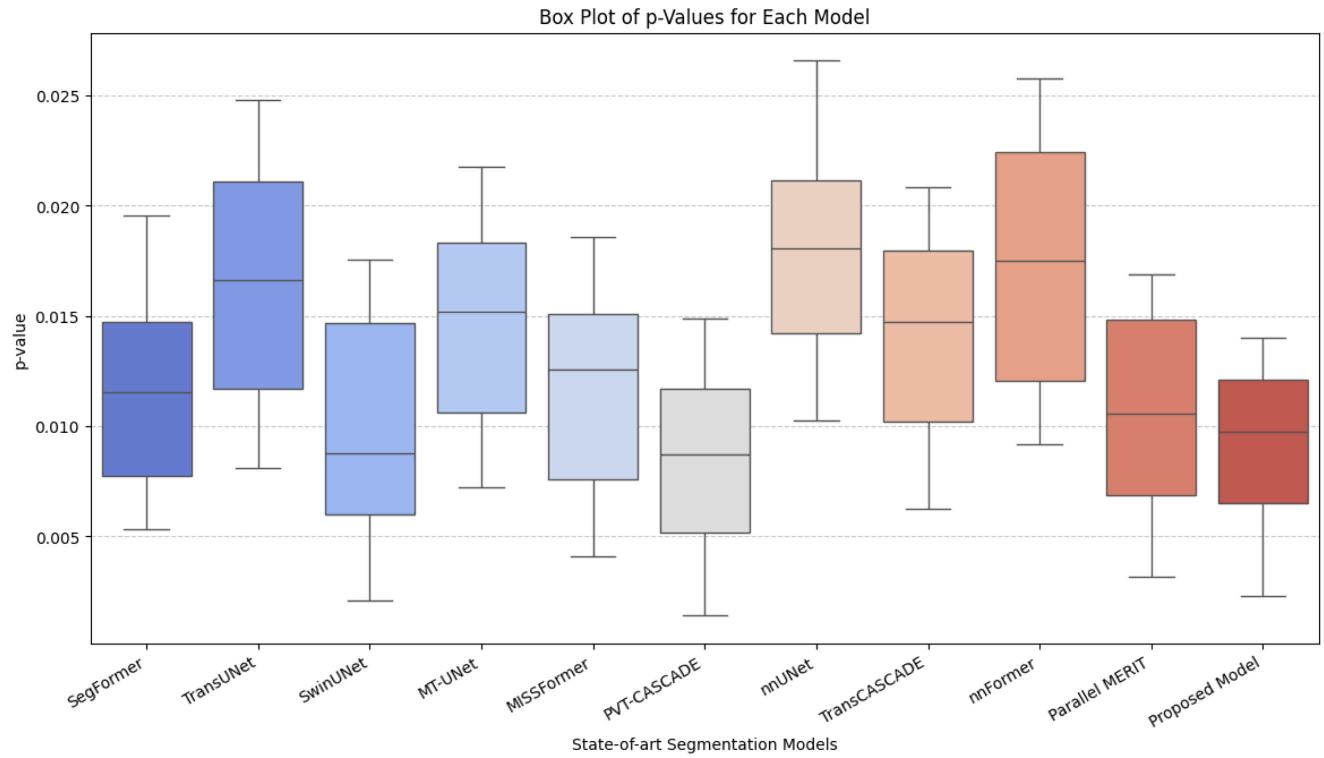

FIGURE 6. Box plot representation of the p-values indicating the statistical significance of performance deviation across various state-of-the-art segmentation models. p-value distributions for each model arise from the hypothesis testing of test images, wherein the hypothesis tends to be rejected with decreasing p-value. The proposed model exhibits persistently low p-values, indicating significant improvement over previously established methods is statistically significant and unlikely due to random chance.

TABLE 4. Model performance comparison by varying the decoder parameters. The last experiment indicates the most accurate segmentation results for the set parameters. For the rest of the experiments, one attribute is changed keeping rest constant to highlight the effect of change of that parameter in performance.

Attention Mixing	MSA Projection	Dilations	Attention Aggregation	Mean DICE	Mean HD
No	Convolution	2,3	Concatenation	87.51	12.7
Yes	linear	2,3	Concatenation	86.11	10.5
Yes	Convolution	1,2	Concatenation	87.06	13.5
Yes	Convolution	2,3	Summation	91.15	10.5
Yes	Convolution	2,3	Concatenation	**94.67**	**8.51**

framework's performance. The results illustrated in Table 5 indicate that combining all attention modules appended with skip connections resulted in the highest segmentation accuracy across all skin layers. This framework also outperformed other configurations with different attention setups, proving the importance of the combined architecture regarding global feature extraction and local boundary refinement.

Multi-head self-attention (MSA) specifically allows the model to establish contextual relationships between distant pixels, thus helping with long-range dependencies. This proves to be very useful as far as segmenting large structures such as background (BKG) and hypodermis (HYP), which are more important on a global level than on a fine detail level. Use of just multi-head self-attention, though, could see a drop in performance concerning boundary-sensitive regions such as the epidermis (EPI) and keratin layer (KER). Since MSA cannot effectively capture local spatial features, therefore edge delineation is very weak and misclassification can happen at tissue boundaries.

Through dilated convolutions keeping spatial resolution but extending the receptive field, shallow-wise convolution (SWC) improves local feature extraction. From the result analysis, it was clear that SWC alone performed notably better in EPI and KER segmentation than MSA alone concerning the two layers' complex structural and sharp transitions inherent within such localized extraction of features, which didn't require wider understanding. Without any mechanism for global dependency incorporation, however, SWC couldn't maintain homogeneity within larger areas, leading to reduced performance in minor cases for BKG and HYP. The spatial squeeze-and-excitation attention module (SSAM) retains features better by emphasizing important spatial regions

TABLE 5. Layer-wise segmentation performance of the proposed framework for various attention modules in convolutional attention mixing decoder. The impact of multi-head self-attention (MSA), shallow-wise convolution (SWC), spatial squeeze and excitation attention (SSAM), and attention-based skip connections on segmentation accuracy attained over different skin layers is analyzed.

Models	BKG	HYP	RET	PAP	EPI	KER
MSA + SWC + SSAM	97.9	96.8	95.7	96.3	90.2	90.7
MSA + SWC	97.6	96.3	95.2	96.0	89.8	90.1
MSA + SSAM	97.4	95.9	94.8	95.7	89.5	89.9
SWC + SSAM	97.1	95.5	94.3	95.2	88.9	89.0
MSA only	96.8	95.1	93.9	94.8	88.5	88.6
SWC only	96.5	94.7	93.5	94.5	88.0	88.2
SSAM only	95.9	93.9	92.8	93.7	87.2	87.4
No Attention	95.5	93.5	92.3	93.2	86.8	87.0
Proposed Model	**98.4**	**97.4**	**96.9**	**97.5**	**91.5**	**91.8**

TABLE 6. Result Comparison of the Proposed framework with the state-of-art baseline research [14]. Model is trained on 10x, 5x, and 2x resolution separately to segment the whole slide input image into Epidermis, Dermis, Hypodermis, and Keratin layers. Overall accuracy, F1-Score, and Recall have been computed to compare the performance.

Methodology	Dataset	F-1 Score	Recall	Accuracy
Simon et. al. [14]		-	-	0.85
Proposed Method	10x	0.94	0.96	**0.95**
Simon et. al. [14]		-	-	0.84
Proposed Method	5x	0.92	0.93	**0.93**
Simon et. al. [14]		-	-	0.81
Proposed Method	2x	0.91	0.91	**0.91**

TABLE 7. Layer-wise accuracy comparison of the proposed framework with the baseline paper, when trained and tested on the dataset (10x, 5x, 2x). Most accurate results have been achieved for the dataset captured at 10× magnification.

Methodology	Data	BKG	HYP	RET	PAP	EPI	KER
Simon et al. [14]	10x	95.00	96.20	70.20	80.80	83.10	84.60
Proposed framework	10x	**99.15**	**98.30**	**98.66**	**95.12**	**91.26**	**92.97**
Simon et al. [14]	5x	93.9	95.5	81.8	78.80	84.9	81.2
DermaTransNet	5x	98.35	97.43	96.38	90.98	88.56	87.74
Simon et al. [14]	2x	95.1	94.4	68.5	77.30	74.6	87.8
DermaTransNet	2x	98.65	98.03	96.22	89.81	84.13	86.86

and recalibrating the activations at channel-level. It greatly enhances the accuracy within structured layers like the reticular (RET) and papillary (PAP) dermis, where complex patterns and multi-scale textures prevail. It dynamically adapts the attention weights, which is especially suited for highlighting tissue boundaries and minimizing redundant feature activations. However, when used alone, it lacks multi-scale receptive fields for broad tissue segmentation, leading to reduced performance in larger structures like BKG and HYP when compared to configurations in conjunction with MSA or SWC.

The integration of MSA, SWC, and SSAM leads to a balanced trade-off between long-range dependencies, local spatial understanding, and dynamic feature recalibration within the framework. This combination enhances reticular (RET) and papillary (PAP) segmentation significantly, where the global and local information balance critically impacts the correct delineation of complex tissue structures. It shows that the greater overall segmentation performance is definitely given by this combination over any of the individual components alone. However, while this configuration gives strong results throughout most layers, challenging boundaries such as epidermis (EPI) and keratin (KER) are still associated with minor inaccuracies, most likely arising from a lack of explicit edge reinforcement along the course of feature propagation in these same layers.

Further incorporation of attention-driven skip connections improves boundary preservation and feature refinement even further to produce the best segmentation quality across layers. The skip connections enable better integration of low-level spatial details from the encoder with high-level contextual features from the decoder; thus, allowing better delineation of fine tissue structures. The highest improvements originate from the epidermis (EPI) and keratin layer (KER), traditionally difficult for segmentation, due to their thin and highly variable structures. Intervening early-stage features into later-stage decisions ensures that fine details of texture and shape are retained, therefore minimizing segmentation errors in regions prone to boundary bleeding or loss of features. Hence, the mechanism alleviates some of the drawbacks presented by configurations based solely on hierarchical attention mixing, where deep feature representations may overshadow fine details.

The ablation study substantiates the conclusion that while each attention mechanism exists, acting rather independently, on segmentation performance, their overall cohesivity leads to a significantly superior outcome. This framework efficiently employs global, local, and adaptive attention mechanisms as per the varying structural complexities of different skin layers, rendering it superior to existing segmentation architectures.

3) RESULT COMPARISON WITH THE BASELINE METHOD

To compare the performance of the proposed model further, it has been compared with the baseline model for segmenting the whole slide image into epidermis, dermis, hypodermis, and keratin using various evaluation metrics on all image resolutions available in the dataset i.e., 10x, 5x, and 2x. Table 6 indicates the F1- score, recall, and accuracy of the proposed model when trained and tested on the entire dataset i.e., 10x, 5x, and 2x resolution. It is evident from the results, that overall accuracy has been significantly increased in the proposed framework as compared to the baseline model.

Since the model extracted the most influential features and textural details from the input image, which contributes most towards accurate skin disease segmentation, therefore despite of significantly reduced number of parameters, the model has resulted in high accuracy. This has also reduced the computational cost since the requires less training time due to convolutional projection and attention-based skip connections.

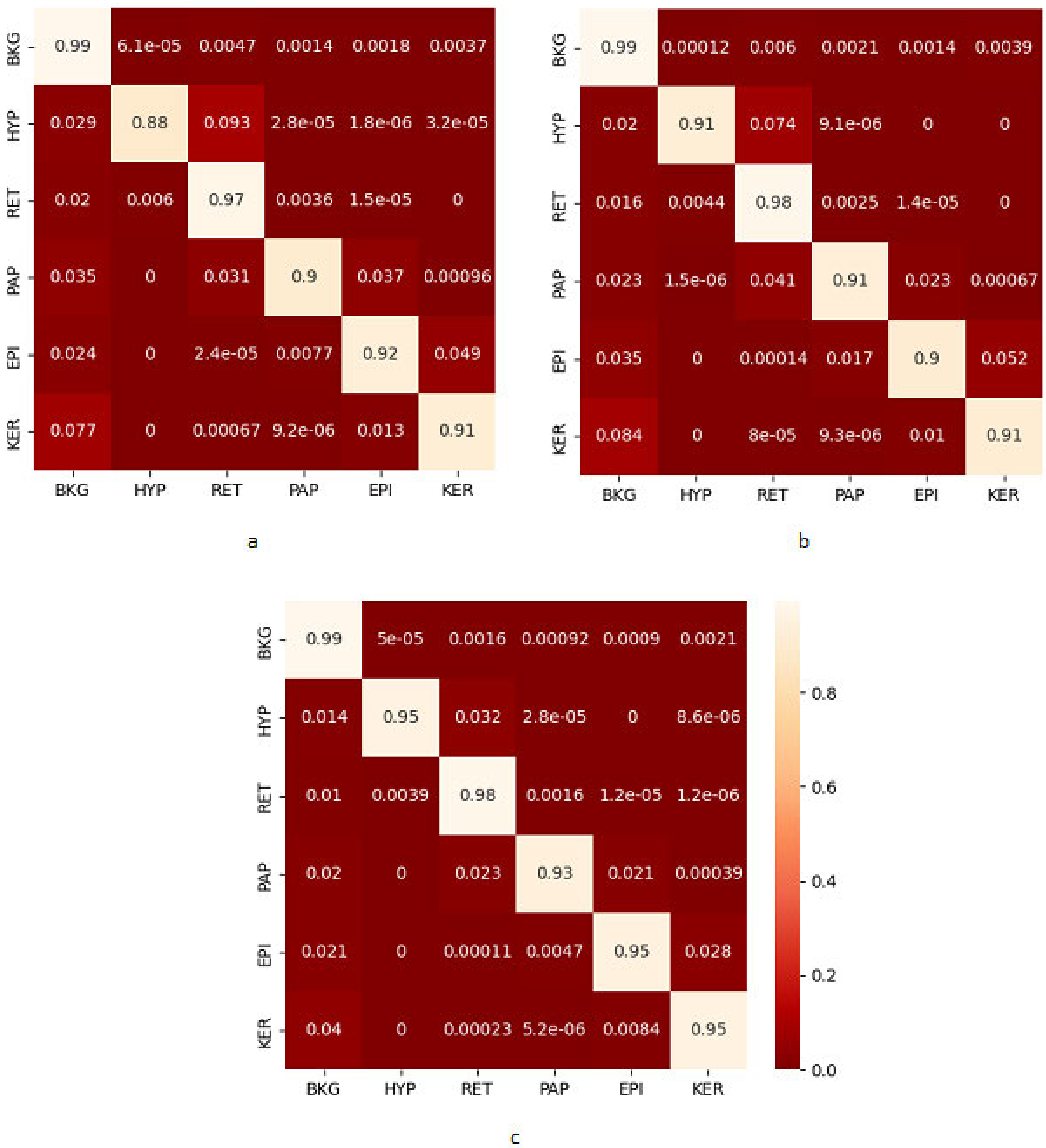

FIGURE 7. Class-wise Confusion Matrix for proposed architecture when trained and tested for classes 0-5, i.e., BKG, HYP, RET, PAP, EPI, KER on images with resolution 2x, 5x, and 10x shown in Figure a-c, respectively.

The resultant model has also been evaluated for each layer separately to calculate the accuracy of each layer, since in medical imaging and autonomous detection systems, individual class accuracy is equally important, as it helps to ensure each class is being extracted accurately. Table 7 indicates the individual segmentation accuracy of each layer achieved through the trained model. The proposed framework yielded a significant improvement in the segmentation accuracy of Keratin, Reticular dermis, Epidermis, and Papillary dermis layers, as compared to the baseline segmentation model proposed in [14]. The attention-based mechanism in encoder module, decoder module, and skip connections has enabled the model to learn the relevant details and has also enabled the model to segment the classes with unequal or having relatively less class-wise representation more accurately. Accurate segmentation of these layers will help in an improved extraction of the desired region for disease-specific classification, which will result in more accurate

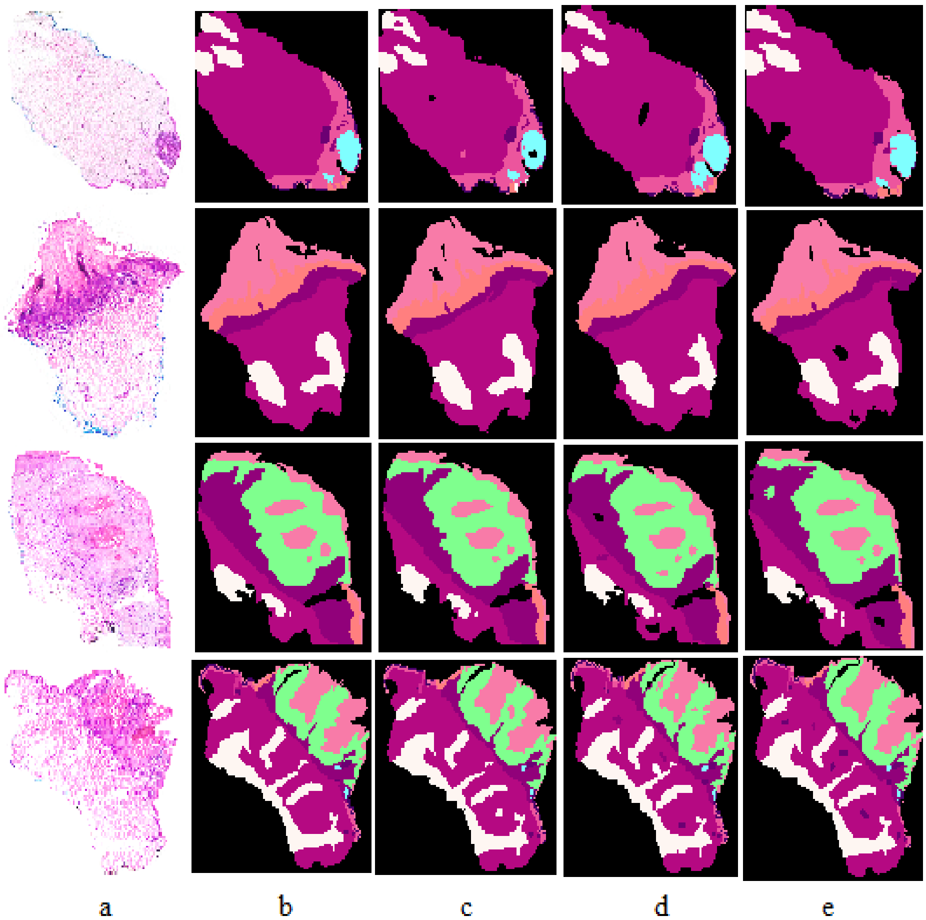

FIGURE 8. Qualitative comparison of results of whole slide images captured at various magnification levels. (a) Whole slide image of skin tissue, (b) Ground truth annotations, (c) Segmentation results at 10× magnification, (d) Segmentation results at 5× magnification, and (e) Segmentation results at 2× magnification. The comparison highlights how segmentation accuracy changes across magnifications.

TABLE 8. Class-wise precision, recall and F1-Score for SegFormer-b0 model when trained and tested on 10x, 5x and 2x respectively.

Resolution	Class	Precision	Recall	F1-Score
10x	0	**0.99**	**0.99**	**0.99**
	1	**0.95**	**0.98**	**0.97**
	2	**0.98**	**0.99**	**0.99**
	3	**0.93**	**0.95**	**0.94**
	4	**0.95**	**0.91**	**0.93**
	5	**0.95**	**0.93**	**0.94**
5x	0	0.97	0.98	0.98
	1	0.91	0.98	0.94
	2	0.98	0.96	0.97
	3	0.91	0.92	0.90
	4	0.90	0.90	0.89
	5	0.91	0.91	0.88
2x	0	0.94	0.97	0.96
	1	0.88	0.97	0.92
	2	0.97	0.94	0.95
	3	0.90	0.90	0.90
	4	0.90	**0.91**	0.87
	5	0.89	0.90	0.88

analysis. It also reduces the computational complexity since instead of evaluating the entire image, extracted layers are processed further for disease classification and analysis.

In medical imaging and computer-aided diagnostic systems, high recall and precision add more reliability to the framework. To further enhance the interpretability of the proposed framework class-wise precision, recall, and F1-Score of the model trained on all magnification levels (10x, 5x, 2x) were computed. As illustrated in Table 8 the model trained and tested for 10x outperformed for all classes (0-(5) as compared to the rest of the models i.e., 5x and 2x, providing a clear insight into the model's performance in terms of the correct segmentation of each pixel to the belonging class. High recall and F1-Score indicate a large proportion of true positives and reduced false negatives, which indicates the model is more predictable and reliable. High recall is crucial in computer-aided diagnostic systems when the consequence of missing positive instances is significant. Conversely, an increased precision indicates that the model has minimized false positives, leading to higher accuracy in classifying instances as positive samples of the class. Moreover, class-wise recall and precision of the model trained and tested on 5x and 2x shows significant improvement compared to the baseline framework [14], indicating an overall increased performance irrespective of the input image resolution.

Furthermore, Figure 7 indicates the detailed class-wise classification accuracy in a confusion matrix, which elaborates a high true positive rate for each class. For all resolutions, the background is the most correctly classified label with the highest true positive rate. Hypodermis classification accuracy has significantly increased as we move from 2x to 10x. Classification of reticular dermis which was the most misclassified label in the baseline model, has significantly improved in all three (2x, 5x, and 10x) trained models. An increase in true positives for the layers has tremendously increased the model's reliance making it more beneficial for autonomous disease detection.

C. TESTING THE PROPOSED FRAMEWORK TO SEGMENT ALL OF THE CLASSES AVAILABLE IN GROUND TRUTH

To compare the proposed framework with the baseline method class-wise, the model was further trained to segment all twelve classes provided in the ground truth, namely: Epidermis, dermis, Hypodermis, Keratin, Inflammation, Glands, Hair Follicles, Basal Cell Carcinoma, Squamous Cell Carcinoma, Intra-epidermal Carcinoma, and Background. As evident from Table 9, proposed framework resulted in the highest accuracy against all the layers, even for the classes with relatively less pixel-wise representation, such as reticular dermis and hair follicles yielding average percentage differences of 22.49% and 28.91% respectively for 10x, 5x, and 2x indicating a high degree of improvement in segmentation of these classes as compared to baseline segmentation model. The proposed multi-resolution, attention-based Encoder-Decoder skip connection architecture has helped the model learn both global and local context thus enabling it to learn intricate textural and structural details at all resolutions.

Figure 8 displays a qualitative comparison of the segmentation outputs measured against the ground truth for various magnifications, i.e., 10×, 5×, and 2×. The results show that the model effectively captures structural details at higher magnifications (10×), with finer segmentation boundaries that closely align with the ground truth. However, at lower magnifications (5× and 2×), some degradation in segmentation quality was observed, leading to imprecise boundary delineation and mild misclassification of certain areas. The failure cases, especially for the 2× magnification, show that fine-grained structure segmentation is hampered by loss of feature representation. These results indicate that while the model generalizes well, the incorporation of some multi-scale information fusion or refinement of the attention mechanisms may certainly increase segmentation performance across the magnifications.

To further elaborate on the performance of the proposed framework, violin plots were generated to visualize class-wise accuracy. The violin plots illustrated in Figure 9 represent the probability distributions of segmentation accuracies for twelve distinct categories segmented by the proposed framework on whole slide images. The clustered values around high probabilities across the majority of categories indicate the robustness and precision of the segmentation framework.

As evident from the plots, categories such as Basal Cell Carcinoma, Reticular Dermis, and Inflammation demonstrate exceptionally compact distributions with median accuracy values surpassing 0.90, indicating a high degree of accuracy in their segmentation. The more extensive distributions observed for categories such as follicles, Squamous Cell Carcinoma, and Intra-epidermal carcinoma imply variability in segmentation, due to inherent complexities in these structures or overlaps in their morphological attributes. Moreover, the occurrence of outliers in specific categories, such as Epidermis, Inflammation, and carcinomas (Squamous and Intra-epidermal), signifies isolated cases of unusual segmentation, which may arise from unique challenges faced in those areas of the WSI. Ultimately, the findings reinforce the framework's proficiency in attaining accurate and consistent segmentation across a diverse range of tissue structures, thereby making it a promising resource for meticulous histopathological examination. In addition to quantitative evaluation, the model has also been evaluated by computing class activation maps as illustrated in Figure 10, where each image has been assessed class-wise to interpret the model's behavior towards each pixel's classification. Class activation maps highlight the significance of each region for the model in making a decision for each class, where red indicates the highest importance in the selected color map and blue indicates the least important features for that class.

In Figure 10, the first column indicates the original histopathological images, second column represents the ground truth annotations, where each color represents a specific tissue type or disease biomarker. The subsequent columns illustrate the predicted probability maps for each of the 12 classes. Each heatmap uses a color scale to illustrate the network's confidence i.e., probability values, where warmer colors indicate high probability in the class prediction and cooler colors reflect low probability or absence of that class. The consistency and clarity of these heatmaps are essential in evaluating the model's ability to accurately differentiate between different tissue types. Notably, in all of the classes, the model's relevant area to get to a class decision accurately maps towards that particular class region, as indicated in ground truth. Additionally, if there is no representation of any of the classes in a certain patch, the model does not highlight any region with high confidence indicating low probability values for that class. These interpretative heatmaps enhances the model's transparency in decision-making and clinical reliability, allowing dermatopathologists to cross-validate AI-driven segmentation with expert knowledge.

D. EVALUATING THE PROPOSED FRAMEWORK ON ANOTHER AVAILABLE DATASET

To evaluate the robustness of the proposed framework, it has been trained and tested on a publicly available dataset of

TABLE 9. Accuracy comparison of the proposed pipeline with the baseline paper [14] for all 12 classes indicated in the ground truth. Highest accuracy has been achieved for 10x resolution. Despite training the model for twelve classes, the model captured details of each class resulting in more accurate segmentation when compared to the baseline model. Highest accuracy has been achieved for hypodermis in the 10x resolution dataset.

Methodology	Data	BKG	HYP	RET	PAP	EPI	KER	INF	GLD	FOL	SCC	BCC	IEC	Overall
Baseline [14]	10x	95.00	96.20	70.20	80.80	83.10	84.60	57.40	87.30	61.50	85.70	86.50	70.70	79.91
Proposed		**98.34**	**97.17**	**96.89**	**96.35**	**90.00**	**91.32**	**93.71**	**92.66**	**90.18**	**92.57**	**97.14**	**93.11**	**94.12**
Difference %		3.34	0.97	26.69	15.55	6.90	6.72	36.31	5.36	28.68	6.87	10.64	22.41	**14.20**
Baseline [14]	5x	93.90	95.50	81.80	78.80	84.90	81.20	61.70	87.40	61.60	78.30	76.80	56.50	78.20
Proposed		93.10	95.78	94.88	89.19	87.71	86.14	86.77	90.13	85.78	86.90	85.73	89.12	89.26
Difference %		0.80	0.28	13.08	15.55	10.39	4.94	25.07	2.73	24.18	8.60	8.93	32.62	12.26
Baseline [14]	2x	95.10	94.40	68.50	77.30	74.60	87.80	64.00	92.10	52.90	75.60	80.70	65.20	77.35
Proposed		98.65	98.03	96.22	89.81	84.13	86.86	87.92	90.15	86.78	89.16	84.18	89.98	90.15
Difference %		3.55	3.63	27.72	12.51	9.53	0.94	23.92	1.95	33.88	13.56	3.48	24.78	13.28

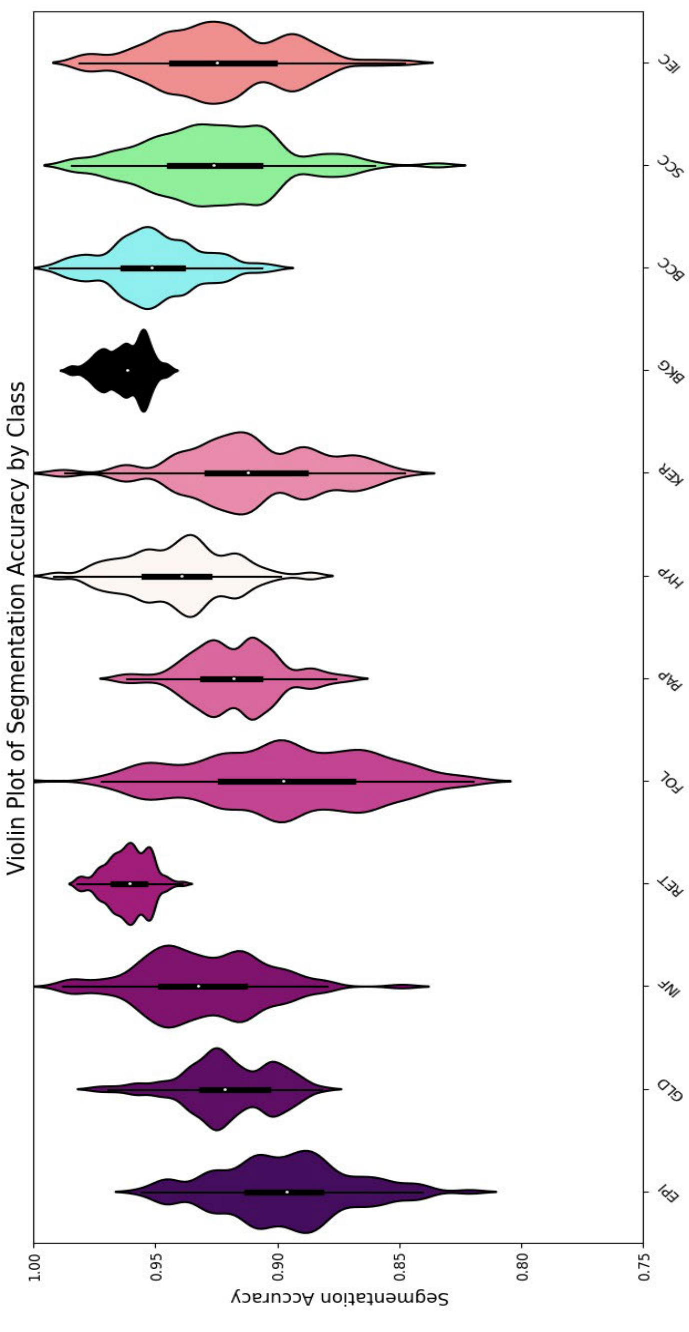

FIGURE 9. The violin plot serves to illustrate the distribution of segmentation accuracy across each of the twelve distinct classes segmented by the proposed segmentation framework. Each individual violin plot exemplifies the variability and distribution of accuracy scores among the test samples corresponding to a particular class. The central black line featured within each plot signifies the median accuracy, whereas the width of the plot at varying points reflects the density of the data.

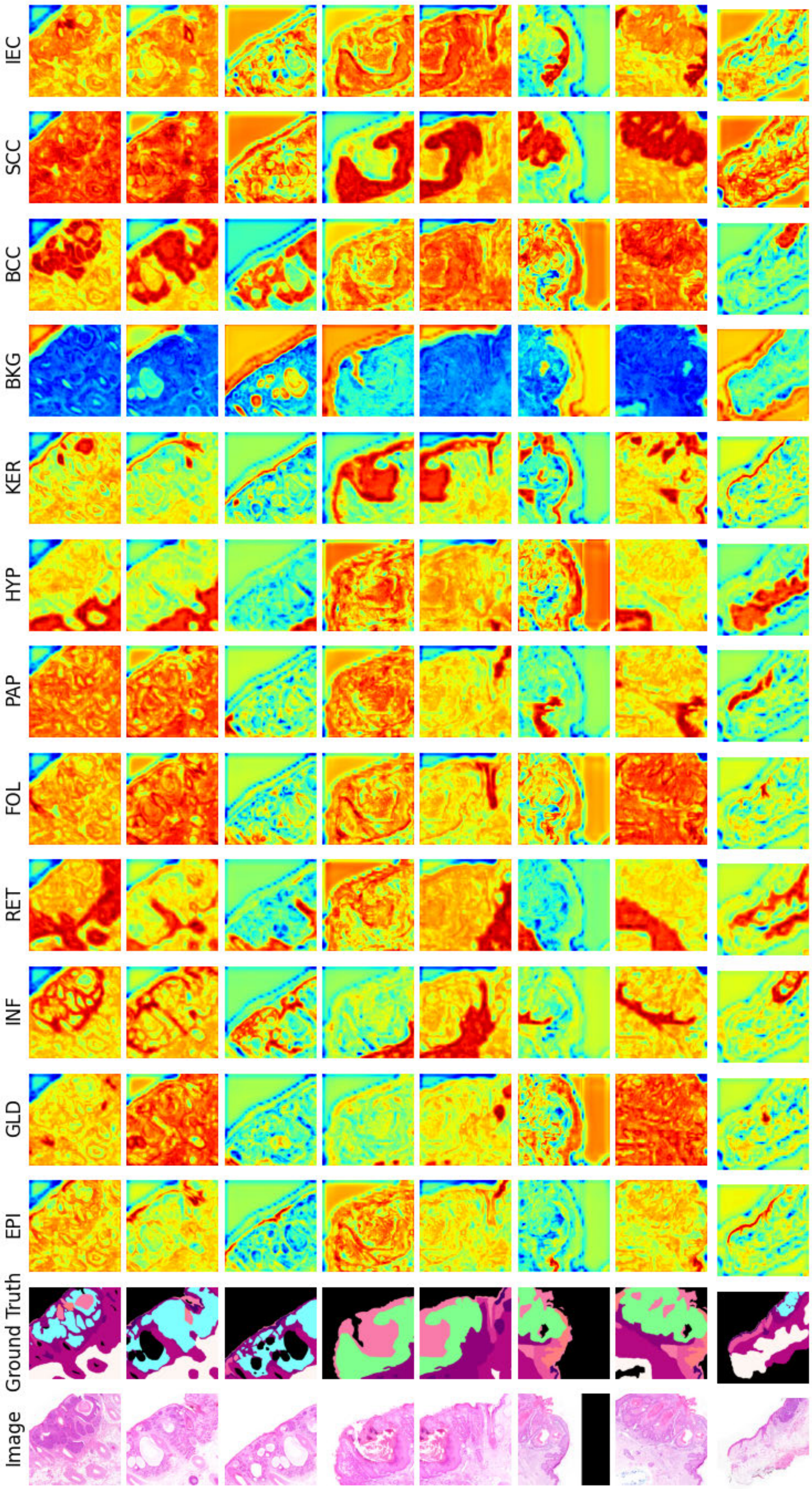

FIGURE 10. Class-wise activation maps for the model trained to segment all classes annotated in the dataset. First column indicates Whole slide Image sample acquired at 10x; Second column indicates the Ground truth of the input samples indicating classes present in each sample; Rest of the columns present Class-wise activation maps to demonstrate model interpretability, where a warmer color represents area of high interest and a cool color indicates area with low interest.

TABLE 10. The framework's performance evaluation was carried out on a dataset comprising H&E-stained skin tissue samples captured at 20x. The proposed framework has outperformed on the tested dataset, indicating robust behavior. Additionally, it establishes the fact that higher-resolution skin tissue samples are better suited for capturing differential tissue structural features.

Dataset	Rate	BKG	HYP	RET	PAP	EPI	KER	INF	GLD	FOL	SCC	BCC	IEC
Queens Land	2×	98.65	98.03	96.22	89.81	84.13	86.86	87.92	90.15	86.78	89.16	84.18	89.98
	5×	93.10	95.78	94.88	89.19	87.71	86.14	86.77	90.13	85.78	86.90	85.73	89.12
	10×	98.34	97.17	96.89	96.35	90.00	91.32	93.71	92.66	90.18	92.57	97.14	93.11
H&E Stained	20×	**98.85**	**97.91**	**97.40**	**97.21**	**92.15**	**92.80**	**94.52**	**93.89**	**91.74**	**93.21**	**97.98**	**94.25**

H&E-stained skin tissue whole slide image samples annotated for twelve classes including skin layers, skin cancer, and tissue structures. Histo-Seg dataset [48] is comprised of 38 whole slide image samples captured at $20\times$ magnification level, where a train-test split of 80%-20% has been utilized. Table 10 presents a detailed comparison of the performance of the dataset with the Queens Land dataset. High segmentation accuracy for all twelve layers demonstrates robustness of the proposed framework. Additionally, the results indicate improved segmentation accuracy after achieving 92.15% and 92.80% for the epidermis (EPI) and keratin (KER) layers, respectively, which usually present a challenge due to their thin structure and subtle intensity differences. Segmentation performance is better at finer levels because they increase detail precision at $20\times$ magnification and allow the model to better delineate tissue borders, lowering some of these segmentation errors in these regions. Tumor-related regions such as squamous cell carcinoma (SCC), basal cell carcinoma (BCC), and intraepithelial carcinoma (IEC) have greatly improved accuracy levels of 93.21%, 97.98%, and 94.25%, respectively. Such improvements have indicated the advantage the model had from containing extra texture and morphological features that are typical of higher-resolution images. The result would be a more accurate boundary delineation for malignant regions. Additionally, the inflammatory (INF), glandular (GLD), and follicular (FOL) layers yield a performance boost of between 1% and 3% compared with the $10\times$ dataset, thus indicating its promise of increased accuracy resolution in the observation of finer patterns attainable from tissue structures. The outcome proves that training as well as testing on the $20\times$ dataset results in significant improvements in segmentation performance for all layers and across pathological regions, due to enhanced spatial resolution features representation, tissue differentiation, and tumor detection. Higher-magnification datasets are therefore best suited for histopathological segmentation and other applications that require fine-grained extraction of features.

V. CONCLUSION AND FUTURE WORK

The proposed deep learning framework semantically segments the input WSI sample into skin layers significant towards disease detection and classification i.e., dermis, epidermis, and hypodermis. In the proposed methodology, instead of segmenting all tissues, layers, and diseases from the input WSI sample, extracting layers in the initial phase has increased class-wise segmentation accuracy compared to the baseline paper. Since all the skin anomalies and diseases lie within these layers, accurate segmentation of these layers significantly increases the disease classification and segmentation accuracy. Thus instead of processing the whole input image, the desired layer can be extracted and passed to the model for disease-specific classification, segmentation, and analysis. After validation employing curated datasets, the early tests done with dermatopathologists confirmed the potential of this model to make diagnostic routines more streamlined. In terms of clinical impact, the proposed framework has a huge influence on dermatopathology because its exact localization of different skin layers and pathological regions is essential for an accurate diagnosis: improved segmentation would serve to enhance diagnostic reliability by providing dermatopathologists-related information regarding tissue structure that is much more consistent and detailed. This could prove especially useful in complex cases where subtle skin layer changes may indicate the presence of the disease, such as a fungal infection or an early-onset malignancy situated either in the epidermis or in the dermis.

By accurately demarcating the most important skin structure including pathological regions, the model also helps dermatopathologists reduce the variability of diagnosis with minimal or even no loss of essential details. For example, the very high accuracy of segmentation with a 20x magnification enables a reliable examination of the dermal-epidermal junction, which is often considered a relevant layer at the beginning phases of evaluating melanoma. The model enables an inner morphometric assessment of dermal structure available at epidermal microscopic levels with high magnification to locate potential fungus presence in the keratin layer, making it possible to diagnose simple superficial fungal infections that are difficult to identify under conventional workflows.

Despite the promising results shown by the proposed methods, several limitations need to be pointed out. To begin with, the model is sensitive to variations in staining, which might compromise its generalizability to datasets prepared using different protocols or various intensities of staining. Since histopathological staining can vary widely between laboratories and even between batches, this dependence could constrain the model's robustness when applied to slides outside the training domain. Furthermore, our proposed framework is trained on H&E stained whole slide image samples. Where, in histopathology, Periodic Acid Schiff stain and immunohistochemistry stain are also being used for disease analysis. To analyse and segment any other stain, the model must be trained on the desired whole slide images. Additionally, as evident from the results proposed framework gives more accurate results for whole slide images captured at high magnification levels. Since, the images captured at high magnification hold more information as compared to the lower ones, thus it helped the model acquire more deterministic features to differentiate each layer more accurately.

This work can be extended by processing the segmented layer for further anomaly detection for example Keratin layer can be further examined for the presence of fungal infections, Hyperkeratosis, and Actinic Keratosis. Similarly, Epidermis can be analyzed for Psoriasis, Eczema, and the presence of Melanoma. Likewise, Dermis can be looked for inflammation and Vascular Abnormalities. Lastly, accurate segmentation of hypodermis will aid in analyzing Lipomas, Liposarcoma, and Cellulitis. Moreover, instead of training the model separately for each magnification level, incremental learning and knowledge distillation based approach can be used to help the model learn from all magnification levels (lower to higher) cumulatively. This will add to the accuracy as instead of relying on one magnification model will extract features from all magnification levels i.e., coarse to fine features.

DATA AVAILABILITY

The dataset used in this research is publicly available as "Histopathology Non-Melanoma Skin Cancer Segmentation Dataset" at https://espace.library.uq.edu.au/view/UQ:8be4bd0.

CODE AVAILABILITY

The implementation of the proposed method, along with necessary training and inference scripts, is publicly available at https://github.com/anumabdulsalam/TransUNET.

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